

SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

TEST CASE DESIGN DOCUMENT
CO4 – Assessment Tool - 1

“Smart Offshore Wind Farm Management Platform”

Submitted by
MANNE PRANEETH KUMAR
Register Number: [192372310]
Course Code: CSA1007
Course Name: SOFTWARE ENGINEERING
Faculty In-Charge: Subramanian. P

Department of Computer Science and Engineering
SIMATS Engineering (Saveetha School of Engineering)
Chennai – 602105, Tamil Nadu
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CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA1007	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	CO4 - Assessment Tool - 1
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics.		
Student Name:	[Student Name]	Reg. No.:	[Register Number]
Date:		Duration:	60 minutes

Code of Conduct

I M.Praneeth/(192372310) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Manne Praneeth

1. Problem Overview

This document addresses Problem Statement 11 — Smart Offshore Wind Farm Management Platform. Offshore wind installations operate in remote, harsh-weather locations where turbines are difficult and expensive to access. Without a centralised system, operators typically discover problems only after a failure occurs, maintenance is planned reactively, and losses caused by weather or under-performing turbines are hard to quantify.

The proposed platform addresses this by bringing turbine telemetry, predictive-failure indicators, weather data, and maintenance workflows into a single operational dashboard, so that plant operators can move from reacting to failures toward anticipating and planning around them. This test case design exercise verifies the correctness, functionality, reliability, and usability of that proposed system ahead of implementation.

Objectives: Monitor turbine performance in real time. Predict equipment failures before they occur. Optimize energy generation. Track weather conditions affecting turbine operations. Schedule maintenance activities efficiently. Generate operational analytics for data-driven decision-making.

Expected Outcomes: Reduced turbine downtime. Improved energy production. Lower maintenance costs. Enhanced grid reliability. Better asset utilization.

2. Functional and Non-Functional Requirements

The requirements below were derived from the six stated objectives of the Smart Offshore Wind Farm Management Platform and form the basis for the test scenarios and test cases that follow.

Table 2.1: Functional Requirements

ID	Requirement
FR-01	The system shall monitor and display real-time turbine performance parameters (RPM, power output, vibration, temperature) for each turbine.
FR-02	The system shall analyse sensor trends and predict potential equipment failures before they occur.
FR-03	The system shall recommend turbine operating settings (e.g., blade pitch, yaw) that optimise energy generation for current wind conditions.
FR-04	The system shall track weather conditions (wind speed, wave height, visibility, temperature) at each site alongside turbine data.

ID	Requirement
FR-05	The system shall allow operators to create, update, and track maintenance activities for individual turbines.
FR-06	The system shall generate operational analytics reports (downtime, output, maintenance history) for a user-selected date range.
FR-07	The system shall raise an alert when a monitored parameter crosses a defined safety or performance threshold.
FR-08	The system shall restrict maintenance-scheduling and configuration actions to authorised operator/administrator roles.

Table 2.2: Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR-01	Reliability	The monitoring service shall maintain at least 99.5% uptime.
NFR-02	Performance	The dashboard shall reflect new telemetry within 5 seconds of receipt.
NFR-03	Usability	Critical alerts shall be visible within two clicks/taps from the dashboard home screen.
NFR-04	Security	Sensor and maintenance data shall be encrypted in transit and at rest, with role-based access control enforced on all write actions.
NFR-05	Scalability	The platform shall support at least 200 turbines reporting concurrently without degradation in refresh performance.
NFR-06	Resilience	The platform shall tolerate intermittent connectivity loss from offshore turbines without losing or corrupting historical data.

3. Test Scenarios

Eight test scenarios were identified to give complete coverage of the requirement set above, one per major platform capability.

Table 3.1: Test Scenarios

ID	Scenario Description	Requirements Covered
TS-01	Real-time turbine performance monitoring and dashboard accuracy	FR-01, NFR-02
TS-02	Predictive failure alert generation and threshold handling	FR-02, FR-07, NFR-05
TS-03	Energy-generation optimisation across the operating wind-speed range	FR-03
TS-04	Weather data tracking and severe-weather handling	FR-04

ID	Scenario Description	Requirements Covered
TS-05	Maintenance activity scheduling and conflict handling	FR-05
TS-06	Operational analytics report generation for a selected date range	FR-06
TS-07	System resilience to invalid sensor data and connectivity loss	NFR-01, NFR-06
TS-08	Role-based access control for maintenance and configuration actions	FR-08, NFR-04

4. Test Design Techniques Applied

Four complementary test design techniques were applied so that the test cases in Section 5 exercise the system's input domains, boundaries, condition combinations, and workflow states with justified, non-redundant coverage.

4.1 Equivalence Partitioning

Input parameters were divided into valid and invalid classes so that one representative value per class stands in for the whole class.

Table 4.1: Equivalence Classes

Input Parameter	Valid Class	Invalid Class
Wind Speed (m/s)	3 – 25 (normal range), e.g. 12	< 3 or > 25, e.g. -5 or 150
Vibration Reading (mm/s)	0 – 4.4 (normal), e.g. 2.0	≥ 4.5 (alert range) or negative, e.g. 5.0 or -1
Maintenance Task Date	Today or a future date	A past date
User Role	Operator / Administrator (authorised)	Viewer / unauthenticated (unauthorised)

4.2 Boundary Value Analysis

Values were tested just below, at, and just above each operational boundary, since defects cluster at these edges.

Table 4.2: Boundary Values

Parameter	Boundary	Values Tested (Below / At / Above)
Wind speed cut-in (m/s)	3	2.9 / 3.0 / 3.1
Wind speed cut-out (m/s)	25	24.9 / 25.0 / 25.1
Vibration alert threshold (mm/s)	4.5	4.4 / 4.5 / 4.6

Parameter	Boundary	Values Tested (Below / At / Above)
Analytics report date range	Single-day range	start = end (1-day) / start = end+1 (invalid)

4.3 Decision Table Testing

A decision table combines the three key turbine health conditions to determine the correct alerting/shutdown action, ensuring every meaningful combination is tested rather than each condition in isolation.

Table 4.3: Turbine Alert Decision Table

Rule	High Vibration (≥ 4.5 mm/s)	High Temp ($\geq 60^{\circ}\text{C}$)	RPM Out of Range	Resulting Action
R1	N	N	N	Normal operation — no alert
R2	Y	N	N	Predictive Warning — vibration
R3	N	Y	N	Predictive Warning — temperature
R4	Y	Y	N	Critical Alert — maintenance auto-suggested
R5	N	N	Y	Critical Alert — RPM fault
R6	Y	Y	Y	Emergency Shutdown — Critical Alert + urgent maintenance auto-scheduled

4.4 State Transition Testing

The maintenance task lifecycle is a finite state machine; transition testing confirms every legal transition succeeds and every illegal transition is blocked.

Table 4.4: Maintenance Task State Transitions

Current State	Event	Next State
Scheduled	Operator starts task	In Progress
In Progress	Operator completes task	Completed
In Progress	Operator cancels task	Cancelled
Scheduled	Operator cancels task	Cancelled
Completed	(any)	No transition allowed — terminal state

Current State	Event	Next State
Cancelled	(any)	No transition allowed — terminal state

5. Detailed Test Cases

The test cases below cover normal, boundary, invalid, and exceptional conditions for every scenario in Section 3, using the techniques justified in Section 4. As this is a pre-implementation design document, the Actual Result and Pass/Fail Status columns are marked “Not Run / Pending”; they are completed once the corresponding system build (tracked separately) is run against this design.

5.1 Turbine Performance Monitoring (TS-01)

Table 5.1: Test Cases — Turbine Performance Monitoring (TS-01)

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-01	TS-01	1. Log in as Operator. 2. Open Turbine Monitoring dashboard. 3. Select turbine WT-014 (online). 4. Observe displayed metrics.	WT-014; live feed RPM 15.2, Power 2.1 MW, Vibration 0.8 mm/s, Temp 42°C	Dashboard shows RPM, power, vibration and temperature matching the incoming feed.	Not Run	Pending
TC-02	TS-01	1. Open dashboard for WT-014. 2. Wait for the next telemetry update. 3. Observe on-screen values.	Updated feed: RPM 15.8, Power 2.3 MW	Displayed values update to the new reading within 5 seconds without manual refresh (NFR-02).	Not Run	Pending
TC-03	TS-01	1. Select turbine WT-021. 2. Simulate loss of sensor connectivity for over 30 seconds. 3. Observe dashboard state.	WT-021; connection dropped >30s	WT-021 is flagged “Offline / No Data” with last-known values greyed out and a last-contact timestamp.	Not Run	Pending
TC-04	TS-01	1. Inject a corrupted telemetry packet for WT-014. 2. Observe dashboard response.	RPM = -999 (invalid)	Invalid reading is rejected/flagged; last valid value is retained and a data-quality warning is logged.	Not Run	Pending

5.2 Predictive Failure Alerts (TS-02)

Table 5.2: Test Cases — Predictive Failure Alerts (TS-02)

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-05	TS-02	1. Set vibration alert threshold to 4.5 mm/s. 2. Feed a vibration reading of 4.5 mm/s for WT-009. 3. Check Alerts panel.	Vibration = 4.5 mm/s (boundary)	A Predictive Failure Warning is raised for WT-009.	Not Run	Pending
TC-06	TS-02	1. Feed a vibration reading of 4.4 mm/s for WT-009. 2. Check Alerts panel.	Vibration = 4.4 mm/s	No alert is raised; turbine status remains Normal.	Not Run	Pending
TC-07	TS-02	1. Simulate a rising vibration/temperature trend over 24h for WT-030. 2. Open the generated alert.	Vibration 2.1→4.8 mm/s; Temp 38→55°C over 24h	Alert lists turbine ID, predicted failure type, a confidence indicator, and a recommended maintenance action.	Not Run	Pending
TC-08	TS-02	1. Simulate threshold breaches on 5 turbines within the same minute. 2. Open Alerts panel.	5 concurrent breaches	All 5 alerts are shown, each attributed to the correct turbine, none dropped or overwritten (NFR-05).	Not Run	Pending

5.3 Energy Generation Optimization (TS-03)

Table 5.3: Test Cases — Energy Generation Optimization (TS-03)

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-09	TS-03	1. Input wind speed 12 m/s, direction 220°, for a turbine cluster. 2. Open Optimization panel.	Wind speed 12 m/s	System recommends blade-pitch/yaw settings that maximise output within the rated range.	Not Run	Pending
TC-10	TS-03	1. Set incoming wind speed to 3 m/s for an idle turbine. 2. Observe turbine and optimisation status.	Wind speed = 3 m/s (cut-in boundary)	Turbine transitions Idle → Generating; optimisation recommendations begin.	Not Run	Pending
TC-11	TS-03	1. Set incoming wind speed to 25 m/s. 2. Observe turbine and optimisation status.	Wind speed = 25 m/s (cut-out boundary)	System recommends safety feathering and halts generation; status shows “Safety Stop.”	Not Run	Pending

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-12	TS-03	1. Feed an out-of-range wind-speed reading. 2. Observe optimisation module.	Wind speed = 150 m/s (invalid)	Reading is flagged as a sensor fault, excluded from the optimisation calculation, and logged.	Not Run	Pending

5.4 Weather Tracking (TS-04)

Table 5.4: Test Cases — Weather Tracking (TS-04)

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-13	TS-04	1. Open the Weather module for a site. 2. Observe displayed metrics.	Site: offshore array, Block-B	Wind speed, wave height, visibility and temperature are shown, each timestamped.	Not Run	Pending
TC-14	TS-04	1. Simulate an incoming storm advisory for a site. 2. Observe all connected dashboards.	Storm advisory, gusts 90 km/h	A prominent warning banner appears on every dashboard; the site's maintenance schedule is auto-flagged “weather risk.”	Not Run	Pending
TC-15	TS-04	1. Open Weather module. 2. Select date range 01–07 Aug 2026. 3. View the trend chart.	Range: 01–07 Aug 2026	Chart plots wind speed and wave height for each day, matching the recorded weather log.	Not Run	Pending

5.5 Maintenance Scheduling (TS-05)

Table 5.5: Test Cases — Maintenance Scheduling (TS-05)

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-16	TS-05	1. Open Maintenance module. 2. Click “New Task.” 3. Select WT-014, task “Bearing Inspection,” date 05 Sep 2026. 4. Save.	Turbine WT-014; date 05-Sep-2026	Task is saved with status “Scheduled” and appears on WT-014's maintenance record.	Not Run	Pending
TC-17	TS-05	1. Attempt to schedule a second task for WT-014 that overlaps an existing one. 2. Save.	Overlapping date/time slot	Save is blocked; a conflict warning names the overlapping task.	Not Run	Pending

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-18	TS-05	1. Mark a Scheduled task “Start.” 2. Verify status. 3. Mark it “Complete.” 4. Verify status and timestamp.	Existing scheduled task	Status moves Scheduled → In Progress → Completed, with a completion timestamp recorded at each step.	Not Run	Pending
TC-19	TS-05	1. Attempt to schedule a maintenance task dated in the past. 2. Save.	Task date = 01-Jan-2025	Entry is rejected with a “date cannot be in the past” validation message.	Not Run	Pending

5.6 Operational Analytics (TS-06)

Table 5.6: Test Cases — Operational Analytics (TS-06)

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-20	TS-06	1. Open Analytics module. 2. Select range 01–31 Jul 2026. 3. Generate report.	Range: 01–31 Jul 2026	Report shows downtime, output and maintenance count per turbine for the range, exportable as PDF/CSV.	Not Run	Pending
TC-21	TS-06	1. Select a single-day range (start = end date). 2. Generate report.	Start = End = 15 Aug 2026	Report generates successfully for that single day without error.	Not Run	Pending
TC-22	TS-06	1. Select an end date earlier than the start date. 2. Attempt to generate report.	Start 20-Aug-2026, End 10-Aug-2026	System shows a validation error and does not generate the report.	Not Run	Pending

5.7 System Resilience & Connectivity (TS-07)

Table 5.7: Test Cases — System Resilience & Connectivity (TS-07)

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-23	TS-07	1. Disconnect turbine WT-045 for 10 minutes. 2. Restore the connection. 3. Check the dashboard and history log.	WT-045 offline 10 min, then reconnected	On reconnect, the timeline gap is backfilled from buffered data or clearly marked; no silent data loss (NFR-06).	Not Run	Pending

5.8 Access Control (TS-08)

Table 5.8: Test Cases — Access Control (TS-08)

TC ID	TS ID	Test Steps	Test Data	Expected Result	Actual	Status
TC-24	TS-08	1. Log in as Administrator. 2. Open a pending maintenance schedule change. 3. Approve it.	Role: Administrator	Approve action is available, succeeds, and is logged against the administrator's ID.	Not Run	Pending
TC-25	TS-08	1. Log in as Viewer. 2. Attempt to open “New Task.” 3. Attempt the equivalent API call directly.	Role: Viewer	“New Task” is disabled/hidden in the UI; the direct API call is rejected (403 Forbidden).	Not Run	Pending

6. Requirement Traceability & Test Coverage

The matrix below maps every functional and non-functional requirement from Section 2 to the scenario(s) and test case(s) that exercise it, confirming complete requirement and functional coverage.

Table 6.1: Requirement Traceability Matrix

Req. ID	Requirement Summary	Scenario(s)	Test Case(s)	Coverage
FR-01	Real-time turbine performance monitoring	TS-01	TC-01 – TC-04	Covered
FR-02	Predictive failure detection	TS-02	TC-05 – TC-08	Covered
FR-03	Energy-generation optimisation	TS-03	TC-09 – TC-12	Covered
FR-04	Weather condition tracking	TS-04	TC-13 – TC-15	Covered
FR-05	Maintenance activity scheduling	TS-05	TC-16 – TC-19	Covered
FR-06	Operational analytics reporting	TS-06	TC-20 – TC-22	Covered
FR-07	Threshold-based alerting	TS-02	TC-05, TC-06, TC-08	Covered
FR-08	Role-based action restriction	TS-08	TC-24, TC-25	Covered
NFR-01	Monitoring reliability / uptime	TS-01, TS-07	TC-03, TC-23	Covered
NFR-02	Dashboard refresh performance	TS-01	TC-01, TC-02	Covered
NFR-03	Alert usability	TS-02	TC-07	Covered
NFR-04	Data security & access control	TS-08	TC-24, TC-25	Covered
NFR-05	Concurrent-turbine scalability	TS-02	TC-08	Covered
NFR-06	Connectivity resilience	TS-01, TS-07	TC-03, TC-23	Covered

7. Conclusion

The test case design presented above provides structured coverage of Problem Statement 11 – Smart Offshore Wind Farm Management Platform across its six core objectives. Fourteen functional and non-functional requirements were identified and mapped to eight test scenarios, from which twenty-five detailed test cases were derived using equivalence partitioning, boundary value analysis, decision table testing, and state transition testing. The accompanying traceability matrix confirms that every identified requirement is exercised by at least one test scenario and test case, giving the design comprehensive functional and non-functional coverage ahead of implementation and execution.

8. References

1. ISTQB. (2023). Certified Tester Foundation Level Syllabus. International Software Testing Qualifications Board.
2. Myers, G. J., Sandler, C., & Badgett, T. (2011). The Art of Software Testing (3rd ed.). Wiley.
3. Sommerville, I. (2015). Software Engineering (10th ed.). Pearson Education.